Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Nexgen

Project / Solution Name

NexFlow

Name (Leader)

TODO - HUMAN VERIFICATION - registered leader

Email & Phone (Leader)

TODO - HUMAN VERIFICATION - email and phone

University Affiliation(s)

TODO - HUMAN VERIFICATION - institution(s)

Live Prototype URL

https://orbyte-smart-yield-nexgen.replit.app/

Code Repository URL (public / access granted to judges)

https://github.com/Nexgen-XMUM/Nexflow

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Browser HTML/CSS/JavaScript calls the FastAPI backend. Catalogue JSON supplies product facts; SQLite stores batches, quotes, orders, points and donation records. Pricing and footprint arithmetic are deterministic. Optional DeepSeek generates wording only; the local competition launcher forces a template. Owners approve promotions. Payment, WhatsApp and recipient handover are simulated.

Replit serves a public related deployment, but HTML and language assets differ from this GitHub baseline; parity is not asserted. Standalone impeccable/ uses a local YOLOv8 weight file for still-image inference, with staff correction and local JSON confirmation. It does not update the main ledger. The welcome animation is preset; browser ONNX experiments are separate. RunPod is not used.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python / HTML / CSS / JS	Main implementation languages	PSF / web standards
FastAPI 0.156.4 Pydantic 2.14.1	API and validation	MIT
Starlette 1.9.0 Uvicorn 0.54.0	ASGI routing and server	BSD-3-Clause
SQLite / HTTPX 0.28.1	Ledger / HTTP client	Public domain / BSD-3-Clause
dotenv 1.2.1 / pytest 9.1.1	Config / offline tests	BSD-3-Clause / MIT
Vision / browser libraries	See Appendix A for scope and versions	See Appendix A

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek (optional)	Promo opening / closing; off in competition runner	User API key; billing tier needs confirmation
WhatsApp / payment / logistics	None: simulated in browser or local ledger	Not used
Google Fonts / CDNs	Fonts; Tailwind CDN on login; ONNX CDN in experiment	Public resource delivery; see Appendix A

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Public prototype: https://orbyte-smart-yield-nexgen.replit.app/	Plan / region / credit redemption: HUMAN VERIFICATION
GitHub	Repository / collaboration: Nexgen-XMUM/Nexflow	Public Git repository
RunPod	Not used; no endpoint, no GPU use claimed	None

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
YOLOv8 bread_finetune	Ultralytics; local best.pt	Reported fine-tuned; lineage / training logs need team verification	AGPL-3.0 terms; verify weight provenance
DeepSeek (configurable)	DeepSeek API	Optional remote text generation; local template by default	Provider API terms
ONNX bread export	train_bread/ and web_online/	Browser experiments only; not main ledger	Verify export lineage; runtime MIT

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Bread Detector	Roboflow Universe	Public; URL in Appendix A	CC BY 4.0	Review images
Bakery fixtures	data/; demo runner	Synthetic records	Project fixtures	Demo IDs; verify
Demo images / weights	impeccable/ and train_bread/	Attributed bread dataset samples / model export	See Appendix A	No live customer collection claimed

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No YES - AI-assisted development used.

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT / Codex	Code, integration, review, documentation	Major; exact model versions need confirmation
DeepSeek AI development harness	Code assistance / debugging (team confirmed)	Major; exact tool version needs confirmation
Claude artifact	Browser prototype reference documented in train_bread/README.md	Extent / model version: HUMAN VERIFICATION
Other tools (incl. Replit AI)	No additional AI use evidenced; team must confirm full list	HUMAN VERIFICATION

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members review generated changes and run automated checks. This submission pass ran 285 backend tests and 49 frontend tests successfully, plus real local HTTP checks. Pricing is calculated in code, not delegated to a model. Model wording remains subject to owner review. The team must verify the final declarations and is responsible for generated output.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor-resource entitlement and redemption require team confirmation.

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Public hosting at the declared live URL	Deployment observed; SDC credit redemption not evidenced
RunPod	None	Not used (team decision)
Other sponsor resources	TODO - HUMAN VERIFICATION	Mentoring / credits / other resources need team confirmation

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The workflow can run locally with FastAPI and SQLite. Replit provides public access but is replaceable hosting; local copy avoids an external model dependency. No RunPod service is used. Sponsor-credit redemption and other sponsor resources have not been verified and must be confirmed by the team.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo stores synthetic user IDs, quotes, orders, points and recipient labels. The local competition runner uses a temporary database deleted on process exit; normal service persistence differs. Optional DeepSeek receives selected product names, quantities and clearance prices, not customer contacts or product costs. Uploaded images and confirmations belong to the independent local vision prototype; retention there is manual. No real authentication is implemented. Replit deployment logging, retention and handling of any visitor-entered information require owner confirmation. Do not enter real customer information into this public prototype.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Generated wording may make unsupported promises; the default local template and owner review reduce this risk. YOLO image counts may be wrong and need human correction; the welcome animation is not inference. Training lineage, sample provenance and the competition synthetic-data rule need confirmation. Public Replit files differ from the checked-out baseline. Role views are browser conventions, not authorization. Payments, messages and handovers are simulated. CO2e uses fixed factors and is not certified avoided emissions. No production safety, privacy or commercial-readiness claim is made.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

HUMAN ACTION: review every statement and tick only verified declarations. Licensing, sponsor use and identity details remain unresolved. No declaration boxes have been checked by the agent.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]

Signature and date are reserved for the authorized team leader. Do not submit until all TODO fields, source/asset rights and declarations are reviewed.

NexFlow Technical Declaration Appendix

Appendix A Libraries and scope

Main runtime versions are pinned in requirements.txt: FastAPI 0.156.4 and Pydantic 2.14.1 (MIT), Starlette 1.9.0 and Uvicorn 0.54.0 (BSD-3-Clause), python-dotenv 1.2.1 (BSD-3-Clause), HTTPX 0.28.1 (BSD-3-Clause), pytest 9.1.1 (MIT). Python uses the PSF license; SQLite is public domain. Transitive dependency versions and licenses must be retained with distributions.

Independent vision dependencies

impeccable/requirements.txt pins Ultralytics 8.4.183, FastAPI 0.161.1 and Uvicorn 0.54.0. app.py imports OpenCV and NumPy; Ultralytics uses PyTorch. OpenCV (Apache-2.0 for current versions), NumPy (BSD-3-Clause), PyTorch (BSD-style); exact installed versions were not verified in an isolated vision environment. Install separately; do not mix the two FastAPI pins. No inference or model training was run in this submission pass.

Browser resources

Tailwind CSS CDN is used by frontend/login_signup_page.html (MIT; CDN version unpinned). ONNX Runtime Web 1.17.3 (MIT) is referenced in train_bread/index.html and the web_online backup. Fonts are fetched from Google Fonts; font-family and downloaded font licenses require team review. Root Apache-2.0 does not replace third-party licenses.

Dataset and model attribution

Bread Detector by breaddetector, Roboflow Universe: https://universe.roboflow.com/breaddetector/bread-detector . Publisher page identifies CC BY 4.0. Dataset page metrics are not NexFlow evaluation results. Repository notes attribute sample images and bread_finetune weights to this project; exact dataset version, split, training logs, authorship and modifications require human confirmation. Retain attribution and mark modifications when verified.

Ultralytics and model rights

Ultralytics documents AGPL-3.0 and Enterprise options: https://www.ultralytics.com/license . No Enterprise entitlement was evidenced. Do not assume the project Apache-2.0 license supersedes YOLO obligations. Confirm the precise weight/export origin and applicable license before signing. The team should confirm whether third-party training images are permitted under the Domain 2 synthetic-data rule.

AI development and deployment evidence

User confirmed ChatGPT/Codex and DeepSeek-assisted development. train_bread/README.md links a Claude artifact; tool version and extent remain human-verification items. Replit URL was supplied by the owner and observed live. Ownership dashboard, deployment source commit, plan, region and sponsor-credit redemption were not accessible. RunPod: Not used. No signature, legal affirmation or platform-use evidence is fabricated.